What we did

RL on #230's `<contacts-v1.multi>` checkpoint, with the reward computed over the **sections of a single rollout** rather than over the rollouts of a group.

Five reward designs, ~25 scored checkpoints, all on 8 × A100 via SkyRL: **M-C** each section's contribution to its own rollout's consensus · **M-F** the last section's F1 · **M-B** the best section's F1 (oracle) · **M-BC** best + consensus · **M-FC** last + consensus · **M-0** a zero-LR control.

Why

#208 ran eleven runs across five rewards and none improved consensus R-precision. The mechanism was a **unit mismatch**: the reward scored one rollout, while the metric votes over 100 *independent* rollouts — an object no rollout can see.

Under `<contacts-v1.multi>` one rollout emits ~22 contact sets in a single generation, so a reward on those is computed on the kind of object the metric scores, with credit assignment inside the sequence.

The result

Three arms beat the warm start on every mode, all CIs excluding zero. Best consensus **0.5775** (M-B), best oracle **0.5663** (M-B), best final section **0.5267** (M-C).

The primary criterion is not met. 22 independent plain rollouts read 0.5896. At a larger budget the two draw level — 0.6054 against plain-100's 0.6058. So RL **closed the gap between the multi format and ordinary sampling; it did not open one.**

#208's negative result is a dose-response, not a verdict

Consensus against distance moved: 0.5673 at KL 0, **0.5775 at 0.009**, 0.5529 at 0.031, 0.3969 at 0.486. Every reward helps at small KL and damages at large.

#208 ran its arms at KL 0.06–0.10 and to 3.96 — past the peak on all of them — and its two arms under 0.0015 never moved. **The window it needed lay between the two learning rates it tried.** Two runs at a 3.3× different rate agree to 0.002 at matched KL, so this is distance, not schedule.

Reward design, in three parts

`E[r] = 0` is necessary and not sufficient. M-C's per-section marginal is **366× larger at 1 section than at 22** — it paid the policy to emit a worse answer — while `E[A] = 0` held exactly, because centring is computed *inside* the quantity being gamed.

Every arm moved its candidate count in the direction its own reward pointed. M-C's gradient is strongly negative (collapsed to 1.1 sections); M-B's strongly positive and self-paying (grew to ~26 and held); M-F's nearly flat — and **a weak gradient is not a safe one, it is an unconstrained one**: M-F ran to 259 sections carrying 1.4 contacts each.

Gates written against the failure you have seen will miss the next one. All three original criteria are one-sided; M-F's failure pushed every one *away* from its threshold and none fired.

Two results that outlive the experiment

Selection is dominated by aggregation. An ORACLE selector of one draft reads 0.5646 where simply voting the same drafts reads 0.5750. A final section should synthesise its predecessors, not pick among them.

The spread is the resource. Making candidates more uniform raises mean section F1 from 0.432 to 0.532 and *lowers* best-of-22 — the same trade that defeats every sharpening reward here, appearing in the corpus's own size law.

What next

Fix M-C's scale bug and re-run it — it is the best final-section arm *despite* the bug. The fix is a rollout-level `GRPO(C_i(all))` base (measured scale-correct) plus a **zero-sum** shaping term. The obvious repair — scoring against the causal prefix — was tested and **refuted**.

Nothing further on M-B. Two rates, 13 checkpoints, a smooth peak, agreement to 0.002.

The diversity gap is a corpus question. One rollout's sections cover 658 distinct pairs against 1,065 for 22 independent rollouts.

Figures

`dose_response.png` — 20 checkpoints, five arms, R-precision against distance.

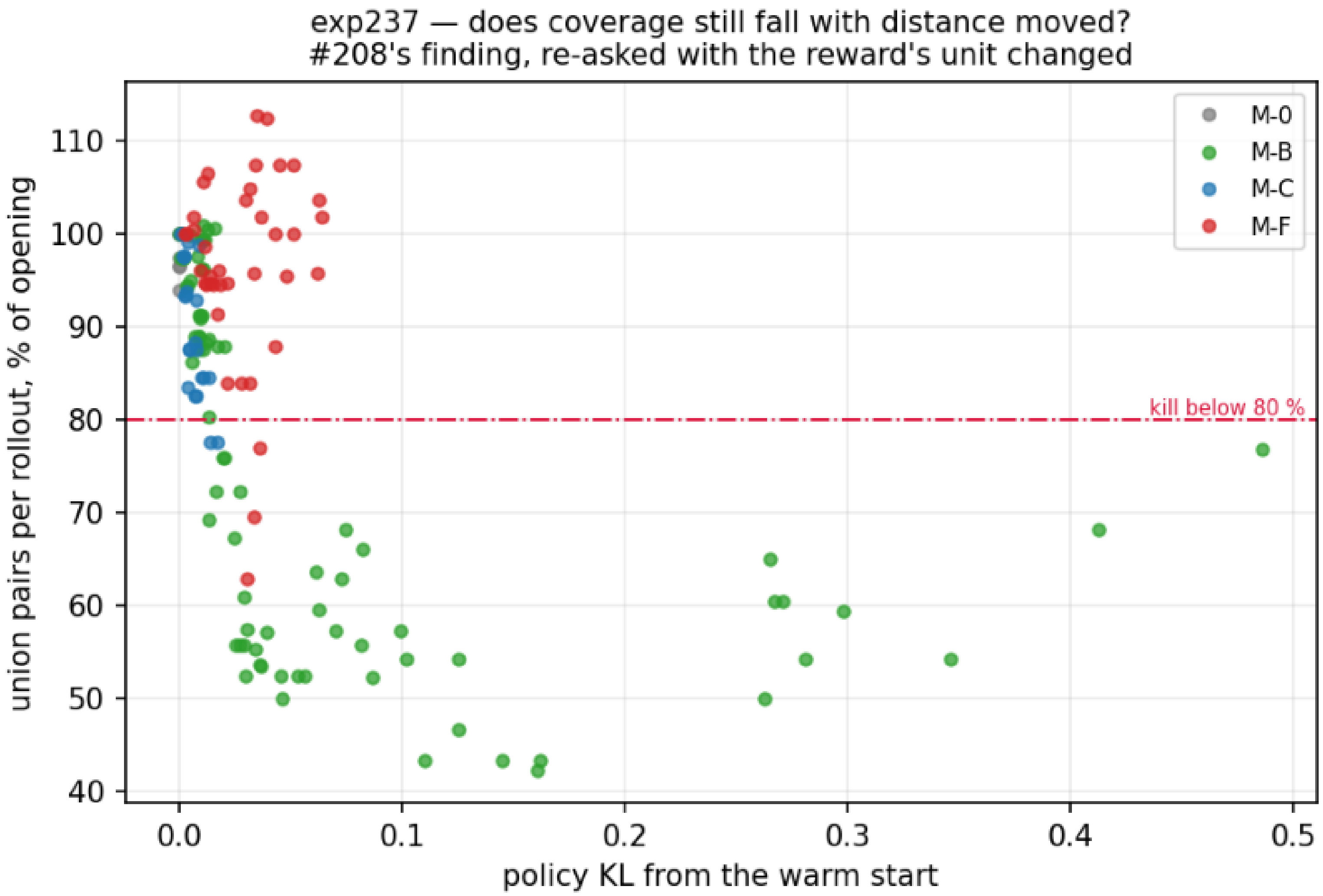
`reward_vs_section_count.png` — each reward's gradient in the count direction.

`section_count_incentive.png` — M-C's scale pathology, 366× at one section.

`gates_over_training.png` · `reward_curves.png` · `reward_shape.png` · `coverage_vs_kl.png`

coverage_vs_kl.png

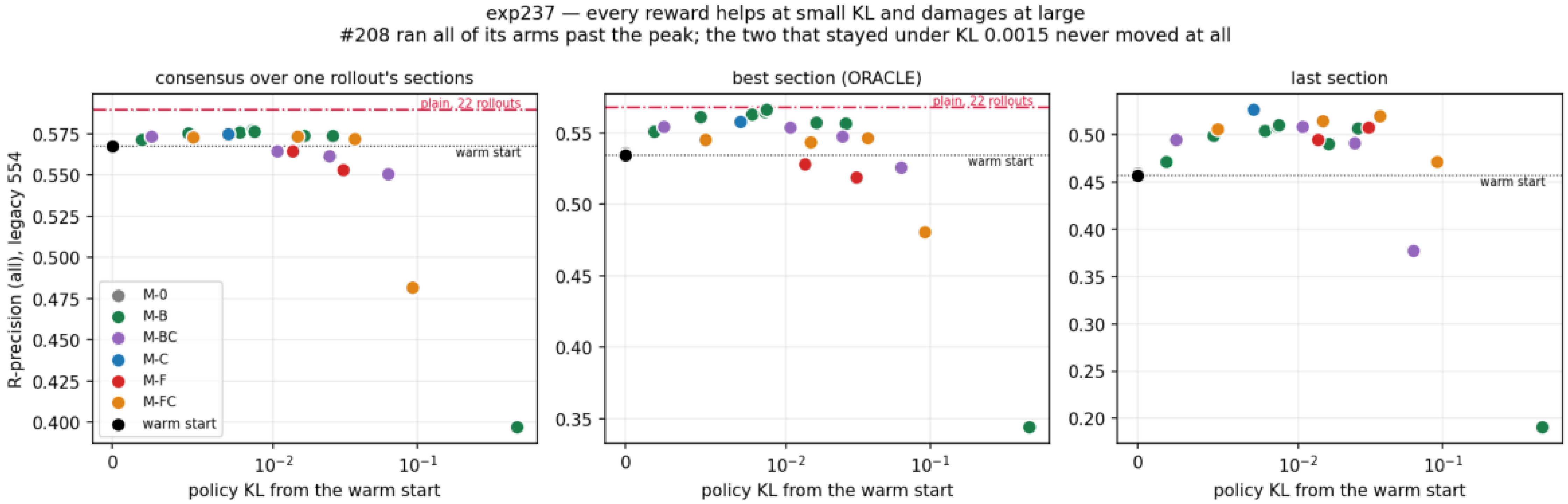
Union coverage against distance moved. M-C crosses the 80% floor by KL 0.013, M-B by 0.016, M-F not until 0.036 -- same destination, different cost per unit of KL.




```
$ make_plots.py --steps data/training_steps.csv.gz --out plots/
```

dose_response.png

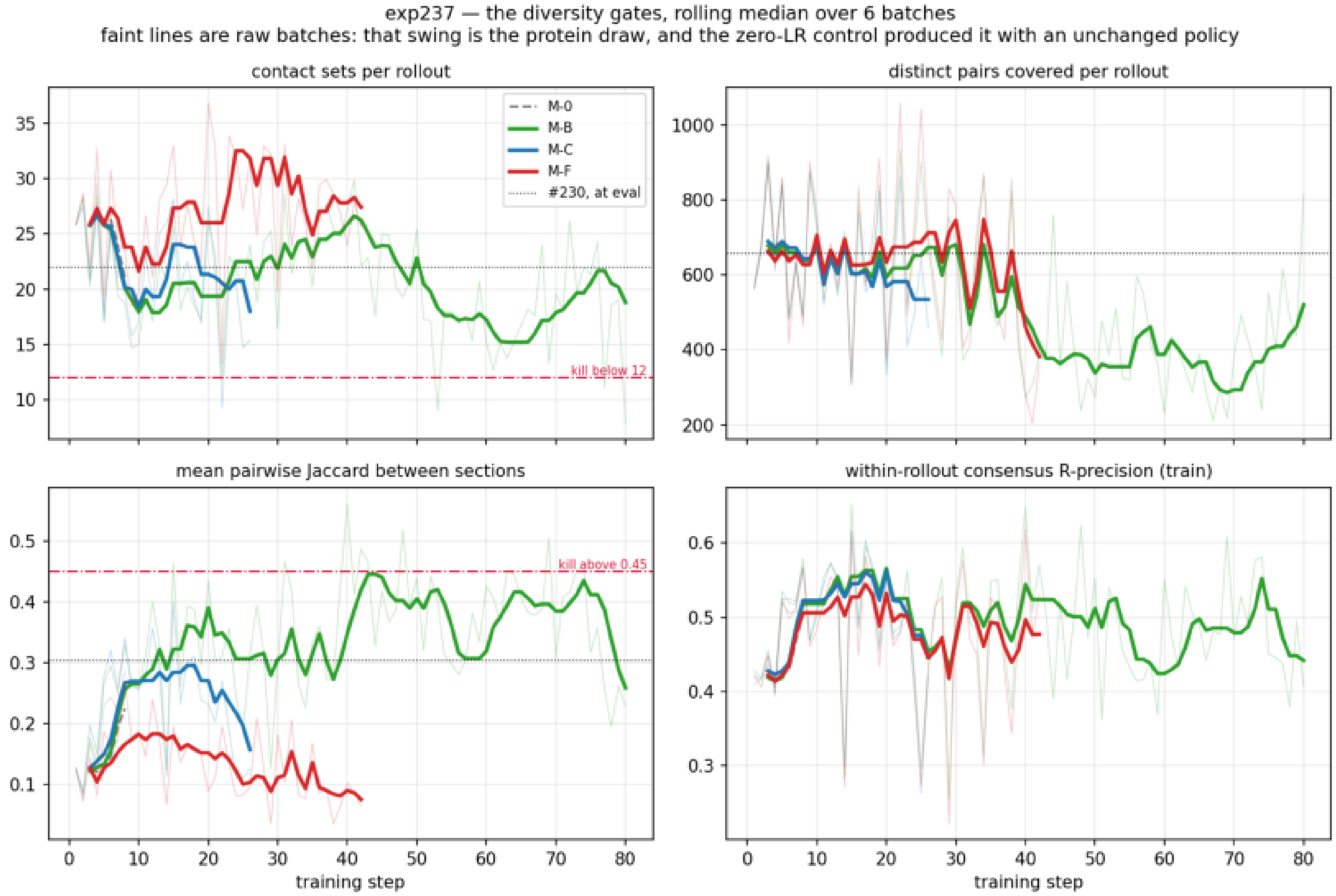
R-precision against distance moved. Consensus peaks at KL 0.007-0.016; #208 ran every arm past it. Red = the budget-matched bar, which nothing reaches.




```
$ plot_reward_shape.py --data data --out plots
```

gates_over_training.png

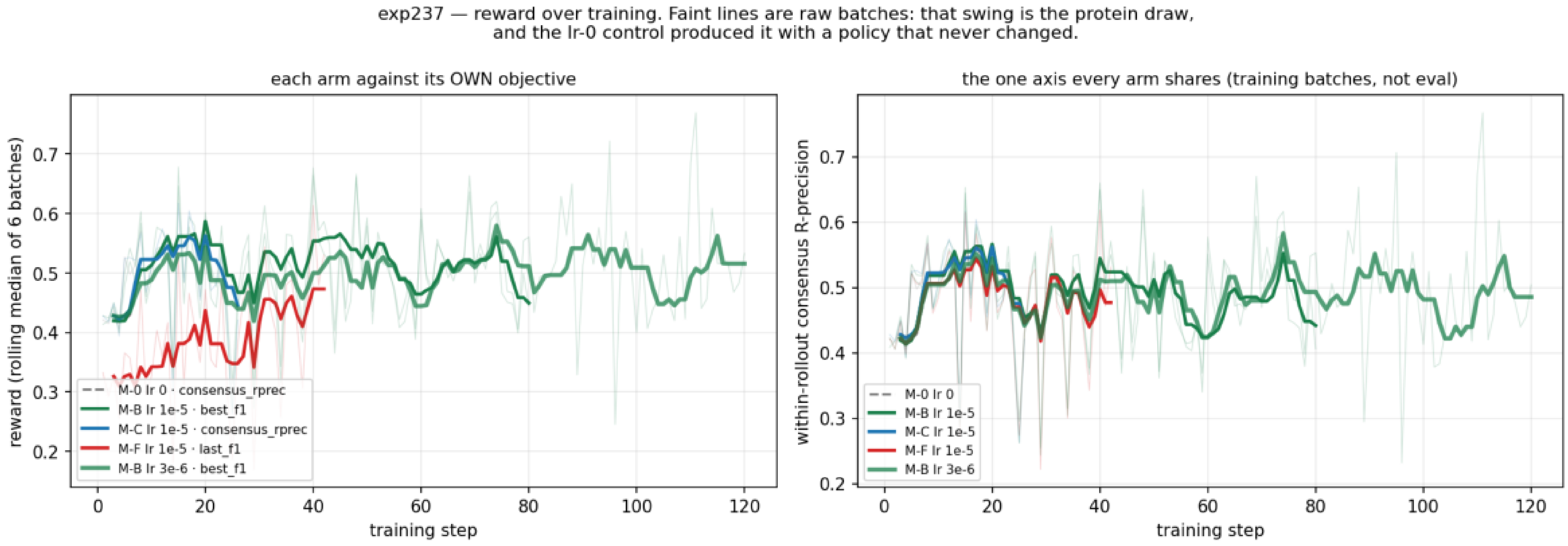
The diversity gates per batch, rolling median over 6. Jaccard separates the arms: M-B rises, M-C and M-F fall. Faint = raw batches, i.e. the protein draw.



`$ make_plots.py --steps data/training_steps.csv.gz --out plots/`

reward_curves.png

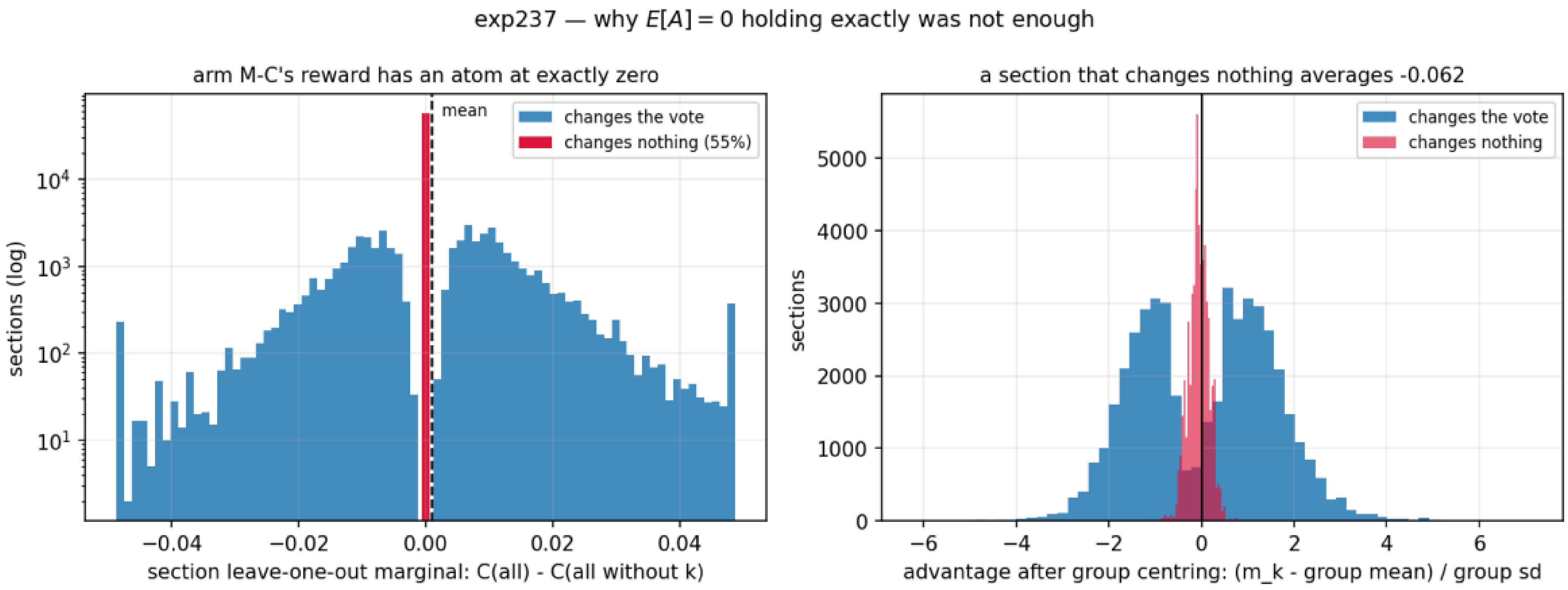
Each arm against its own reward (left) and the shared consensus axis (right), rolling median of 6 batches. Training-batch values on 8 proteins, not eval.




```
$ plot_reward_curves.py --csv data/training_steps.csv.gz --run M-B lr3e-6=/tmp/claude-1000/logs_final/exp237_m_b_lr3e-6.log --out plots
```

reward_shape.png

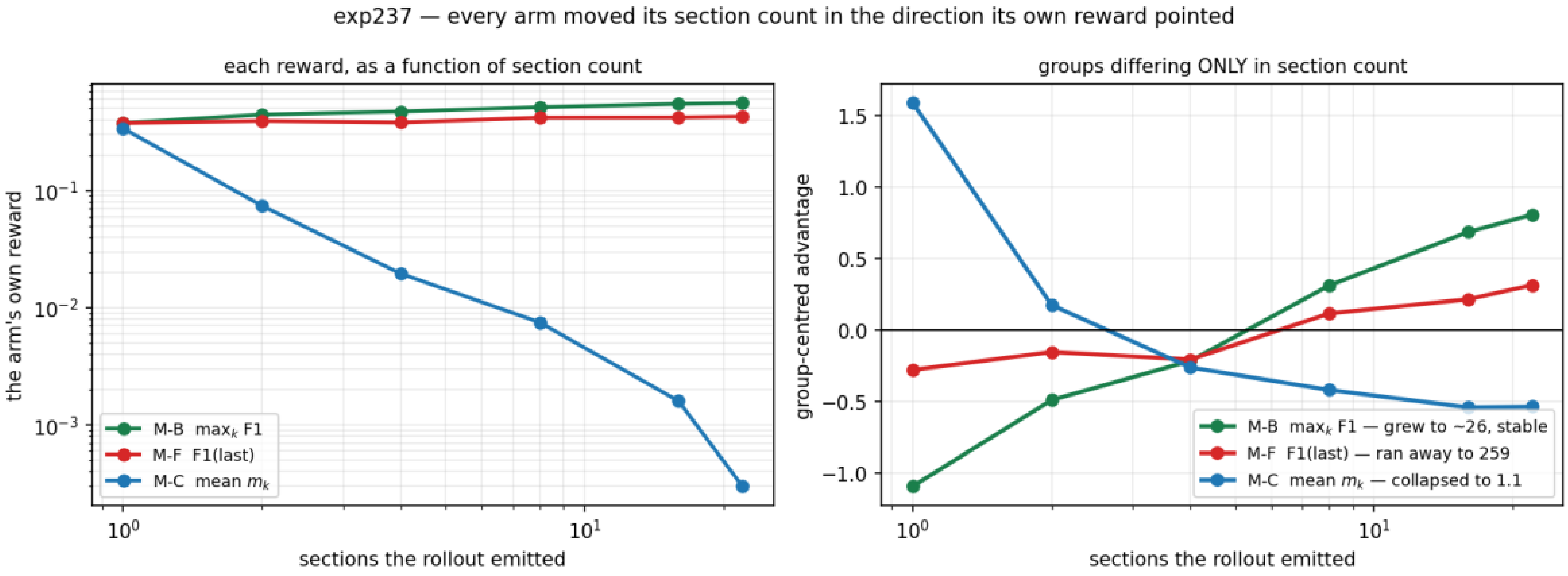
Arm M-C's reward before training: 55% of sections change no vote, and average -0.062 once centred. M-F has no such atom and collapsed the same -- so this is not the cause. See RESULTS.md.




```
$ plot_reward_shape.py --data data --out plots
```

reward_vs_section_count.png

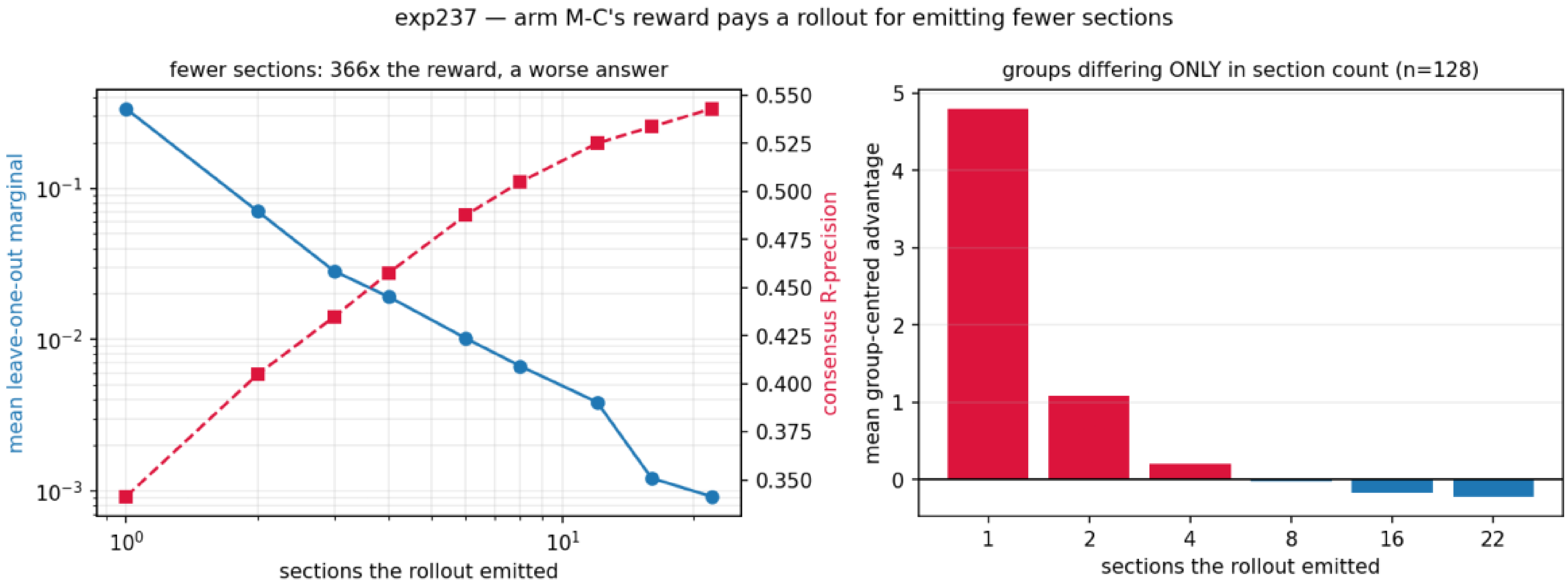
Each reward's gradient in the section-count direction predicts what that arm did: M-C falls (collapsed to 1.1), M-B rises (grew to ~26), M-F is nearly flat (unconstrained, ran to 259).



$ mkfig.py

section_count_incentive.png

m_k is not scale-free in the section count: at 1 section it is 366x its value at 22, while the rollout's own consensus falls 0.543 -> 0.341. Centring does not remove it -- the shorter rollout is the one above its group's mean.



$ plot_reward_shape.py --data data --out plots